ApexPlanet Data Analytics Internship

Task 3: Deep-Dive Analysis Report

Sales Performance, Product Categories, Customer Segments and Monthly Trends

Dataset basis: This report uses the cleaned Task 1 dataset ApexPlanet_Task1_Cleaned_Dataset.csv, not the synthetic sample dataset from the earlier draft.

1. Objective

Build on Tasks 1 and 2 by defining sales KPIs and examining performance by category, product, month, city, age group and gender.

2. Core KPIs

KPI	Result	Formula / rationale
Total Sales	139,399,439.65	Sum of Total_Sales
Unique Orders	1,000	Count of unique Order_ID
Average Sales per Order	139,399.44	Total Sales ÷ unique orders
Total Units Sold	5,435	Sum of Quantity
Repeat Customer Rate (proxy)	5.49%	Customers with 2+ Order_IDs ÷ unique customers × 100

Note: Repeat customer rate is a simple proxy, not a formal cohort-retention or churn metric. Currency was not confirmed, so values are shown without a currency symbol.

3. Deep-dive findings

• **Category:** Electronics has the highest total sales (50,778,581.70); Fashion has the lowest (19,835,895.79).

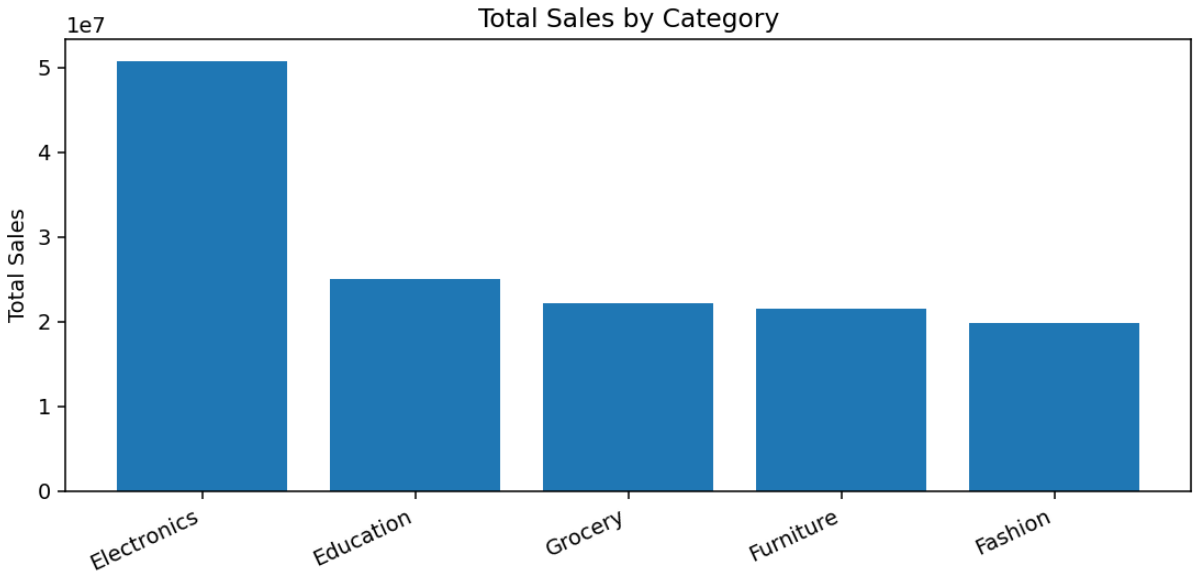
• **Product:** Laptop is the top product by total sales (25,443,008.51).

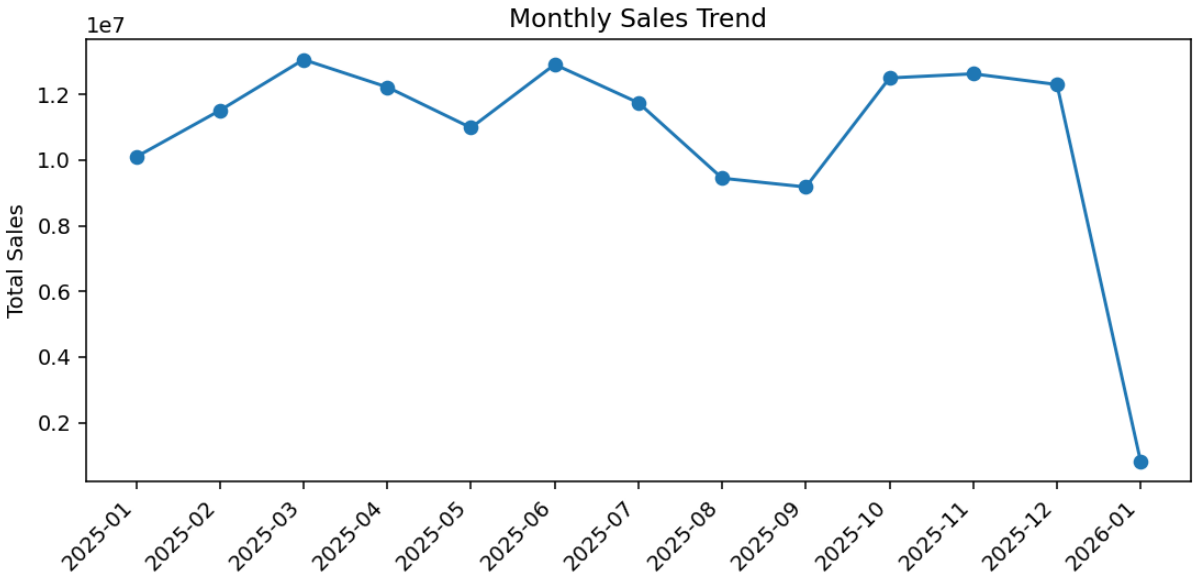
• **City:** Patna has the highest total sales (19,285,966.89).

• **Month:** 2025-03 is the highest-sales month (13,059,899.94).

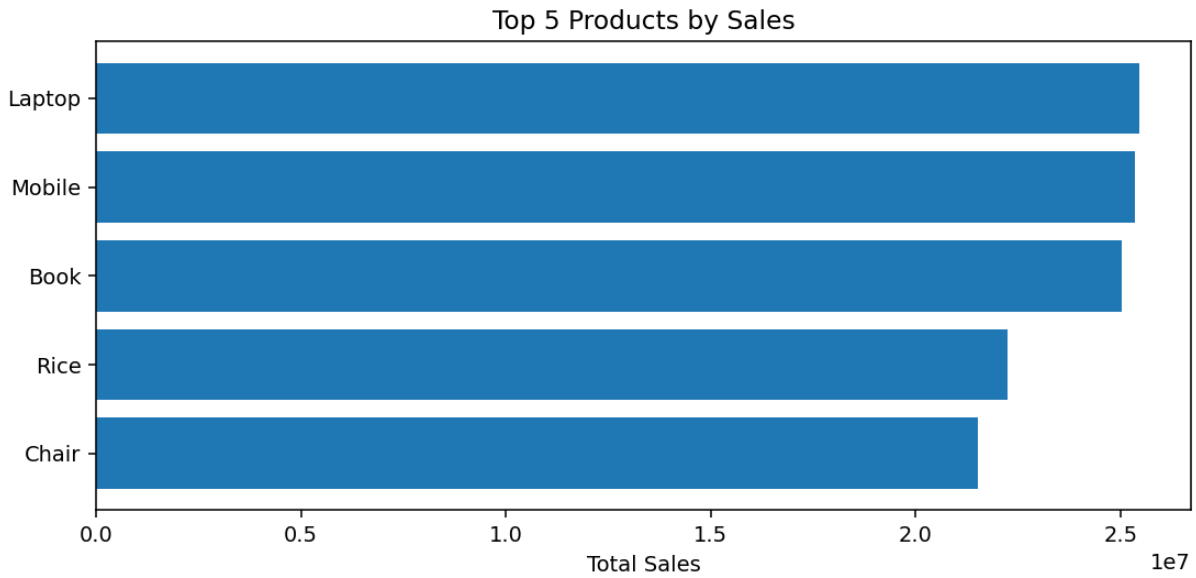
• **Age group:** 36-50 has the highest total sales (45,563,823.42).

• **Gender:** Male has the higher total sales (72,463,550.63) in this dataset.


Total Sales by Category
1e7
5
4
3
2
1
0
Total Sales
Electronics
Education
Grocery
Furniture
Fashion


Monthly Sales Trend
1e7
1.2
1.0
0.8
0.6
0.4
0.2
Total Sales
2025-01
2025-02
2025-03
2025-04
2025-05
2025-06
2025-07
2025-08
2025-09
2025-10
2025-11
2025-12
2026-01

4. Top Product Analysis



5. Business Recommendations

• Investigate the Electronics category to understand which products and customer needs drive sales; consider profit margin before increasing investment.

• Review the lower-performing Fashion category alongside its record count, price range and availability before planning promotions.

• Use monthly sales patterns to plan inventory and campaigns, comparing complete months consistently.

• Check availability and demand for the leading product, Laptop, while considering quantity sold and profitability as well as sales value.

• Use city, age-group and gender patterns to guide further research, not to assume why customers behave differently.

6. Data Quality Notes

Task 1 cleaning notes indicate missing Age values were filled with the median, missing City values were labeled Unknown, repeated Order_ID values were made unique while Original_Order_ID was preserved, sales calculations were checked, and Total_Sales outliers were flagged rather than automatically deleted. Outliers should be reviewed before making changes.

7. Interactive Dashboard Plan

The dashboard should include KPI cards for Total Sales, Unique Orders, Average Sales per Order and Total Units Sold; a monthly sales line chart; category and top-product charts; and filters for date, category, city and age group. The live dashboard link must be obtained by publishing a dashboard online.

Conclusion

This report builds on the cleaned sales dataset and Task 2 analysis themes to identify leading categories, products, cities and months. The published interactive dashboard should use the same dataset and show figures consistent with this report.